

二.必做题 (共1 题,33.3分)

1.

Oppenheim 课本, 第八章课后习题, 8.3, 8.5, 8.12, 8.15, 8.19, 8.23, 8.25, 8.28

答案:

答案解析:

难度: 易

知识点:

三.选做题 (共1 题,33.4分)

1.

下列题目中任选至少 3 道完成

Oppenheim 课本, 第八章课后习题, 8.29, 8.37, 8.42, 8.49, 8.62, 8.64, 8.69

$$\bar{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn}.$$

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- Figure 1 displays three discrete-time signals, $\tilde{x}_1[n]$, $\tilde{x}_2[n]$, and $\tilde{x}_3[n]$, plotted on a horizontal axis representing time n . The signals are shown as vertical stems with dots at the top (positive) or bottom (negative) of the axis.
- $\tilde{x}_1[n]$: Shows a sequence of positive impulses. The first group consists of two impulses, followed by a gap, then three impulses, another gap, three more impulses, a final gap, and three more impulses. Ellipses at both ends indicate the signal continues.
 - $\tilde{x}_2[n]$: Shows a similar sequence of positive impulses as $\tilde{x}_1[n]$, with two, three, three, and three impulses in the respective groups, separated by gaps. Ellipses at both ends indicate the signal continues.
 - $\tilde{x}_3[n]$: Shows a sequence of negative impulses. The first group consists of four impulses, followed by a gap, then four impulses, another gap, four more impulses, a final gap, and four more impulses. Ellipses at both ends indicate the signal continues.

$$IDFT\{X(k)\} = \frac{1}{N} \sum_{k=0}^{N-1} X(k)e^{+j\frac{2\pi}{N}nk} = x[n] \quad 0 \leq n \leq N-1$$
$$X_1(k) = \sum_{n=0}^7 x[n]e^{-j\frac{2\pi}{8}kn} = \sum_{n=0}^7 x[n]e^{-j\frac{\pi}{4}kn} = \sum_{n=0}^3 e^{-j\frac{\pi}{4}kn} = \frac{1 - e^{-j\frac{\pi}{4}k*4}}{1 - e^{-j\frac{\pi}{4}k}} = \frac{1 - (-1)^k}{1 - e^{-j\frac{\pi}{4}k}}$$

$$X_2(k) = \sum_{n=0}^7 x[n]e^{-j\frac{2\pi}{8}kn} = \sum_{n=0}^7 x[n]e^{-j\frac{\pi}{4}kn} = \sum_{n=0}^2 e^{-j\frac{\pi}{4}kn} = \frac{1 - e^{-j\frac{\pi}{4}k*3}}{1 - e^{-j\frac{\pi}{4}k}}$$

$$k = \pm 2 \text{ 非 } 0 \quad k = \pm 4 \text{ 非 } 0 \quad k = \pm 6 \text{ 非 } 0$$

$$X_3(k) = \sum_{n=0}^7 x[n] e^{-j\frac{2\pi}{8}kn} = \sum_{n=0}^3 e^{-j\frac{\pi}{4}kn} - \sum_{n=4}^7 e^{-j\frac{\pi}{4}kn} = \frac{1 - e^{-j\frac{\pi}{4}k \cdot 4}}{1 - e^{-j\frac{\pi}{4}k}} - \frac{e^{-j\frac{\pi}{4}k \cdot 4} - e^{-j\frac{\pi}{4}k \cdot 8}}{1 - e^{-j\frac{\pi}{4}k}}$$

$$X_3(k) = \frac{1 - 2e^{-j\frac{\pi}{4}k \cdot 4} + e^{-j2k\pi}}{1 - e^{-j\frac{\pi}{4}k}} = \frac{2 - 2(-1)^k}{1 - e^{-j\frac{\pi}{4}k}}$$

$$k = \pm 2 \quad \pm 4 \quad \pm 6 \text{ 有: } X_3(k) = 0$$

注意：

这个题 k 的取值不涉及 $e^{-j\frac{\pi}{4}k} = 1$ 的情况，所以可以直接等比数列求和，大家在考试或者做作业过程中一定要注意这里存在的特殊情况，要单独拿出来讨论。

8.5. Compute the DFT of each of the following finite-length sequences considered to be of length N (where N is even):

(a) $x[n] = \delta[n]$,

(b) $x[n] = \delta[n - n_0]$, $0 \leq n_0 \leq N - 1$,

(c) $x[n] = \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \quad 0 \leq n \leq N - 1$,

(d) $x[n] = \begin{cases} 1, & 0 \leq n \leq N/2 - 1, \\ 0, & N/2 \leq n \leq N - 1, \end{cases}$

(e) $x[n] = \begin{cases} a^n, & 0 \leq n \leq N - 1, \\ 0, & \text{otherwise.} \end{cases}$

[解]:

上述所有的序列都视为 N 点序列, 并且明确规定了这里 N 是偶的

(a)

$$x[n] = \delta[n]$$

$$DFT\{x[n]\} = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} = 1 = X(k) \quad k \in [0, N-1]$$

(b)

$$x[n] = \delta[n - n_0] \quad 0 \leq n_0 \leq N - 1$$

$$DFT\{x[n]\} = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} \delta[n - n_0] e^{-j\frac{2\pi}{N}kn} = e^{-j\frac{2\pi}{N}kn_0} = X(k) \quad k \in [0, N-1]$$

(c)

$$x[n] = \begin{cases} 1 & n \text{ even} \\ 0 & n \text{ odd} \end{cases} \quad 0 \leq n \leq N - 1$$

$$DFT\{x[n]\} = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} = \sum_{n \text{ even}} e^{-j\frac{2\pi}{N}kn} = X(k) \quad k \in [0, N-1]$$

我们不妨记 $n \text{ even}$ 为:

$$n = 2m \quad m \in \left[0, \frac{N}{2} - 1\right]$$

Situation1: 如果 $e^{-j\frac{4\pi}{N}k} \neq 1$ 即 $k \neq 0$ or $\frac{N}{2}$

$$X(k) = \sum_{n \text{ even}} e^{-j\frac{2\pi}{N}kn} = \sum_{m=0}^{\frac{N}{2}-1} e^{-j\frac{2\pi}{N}k(2m)} = \sum_{m=0}^{\frac{N}{2}-1} e^{-j\frac{4\pi}{N}mk} = \frac{1 - e^{-j\frac{4\pi}{N}(\frac{N}{2}-1+1)k}}{1 - e^{-j\frac{4\pi}{N}k}} = \frac{1 - e^{-j2k\pi}}{1 - e^{-j\frac{4\pi}{N}k}}$$

$$X(k) = 0$$

Situation2: $k = 0$ or $\frac{N}{2}$ $e^{-j\frac{4\pi}{N}k} = 1$

$$X(k) = \left\{ \left(\frac{N}{2} - 1 \right) - 0 \right\} + 1 = \frac{N}{2}$$

$$X(k) = \begin{cases} \frac{N}{2} & k = 0 \text{ or } \frac{N}{2} \\ 0 & \text{otherwise} \end{cases}$$

(d)

$$x[n] = \begin{cases} 1 & n \in \left[0, \frac{N}{2} - 1\right] \\ 0 & n \in \left[\frac{N}{2}, N - 1\right] \end{cases}$$

$$DFT\{x[n]\} = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}nk} = X(k) \quad k \in [0, N-1]$$

$$X(k) = \sum_{n=0}^{\frac{N}{2}-1} e^{-j\frac{2\pi}{N}nk} = \sum_{n=0}^{\frac{N}{2}-1} \left(e^{-j\frac{2\pi}{N}k}\right)^n$$

Situation1: $e^{-j\frac{2\pi}{N}k} = 1 \quad k = 0$

$$X(k) = \left\{\left(\frac{N}{2} - 1\right) - 1\right\} + 1 = \frac{N}{2}$$

Situation2: $e^{-j\frac{2\pi}{N}k} \neq 1$

$$X(k) = \sum_{n=0}^{\frac{N}{2}-1} \left(e^{-j\frac{2\pi}{N}k}\right)^n = \frac{1 - \left(e^{-j\frac{2\pi}{N}k}\right)^{\frac{N}{2}-1+1}}{1 - e^{-j\frac{2\pi}{N}k}} = \frac{1 - e^{-j\pi k}}{1 - e^{-j\frac{2\pi}{N}k}} = \frac{1 - (-1)^k}{1 - e^{-j\frac{2\pi}{N}k}}$$

$$X(k) = \begin{cases} \frac{N}{2} & k = 0 \\ 0 & k \text{ even (except 0)} \\ \frac{2}{1 - e^{-j\frac{2\pi}{N}k}} & k \text{ odd} \end{cases}$$

(e)

$$x[n] = \begin{cases} a^n & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

$$DFT\{x[n]\} = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}nk} = \sum_{n=0}^{N-1} a^n e^{-j\frac{2\pi}{N}nk} = \sum_{n=0}^{N-1} \left(ae^{-j\frac{2\pi}{N}k}\right)^n = X(k) \quad k \in [0, N-1]$$

这里我们默认 $a \neq 1$ or 0 否则没什么讨论的意义, 这两种情况大家自行判断, 很简单

$$X(k) = \frac{1 - \left(ae^{-j\frac{2\pi}{N}k}\right)^{N-1+1}}{1 - ae^{-j\frac{2\pi}{N}k}} = \frac{1 - \left(ae^{-j\frac{2\pi}{N}k}\right)^N}{1 - ae^{-j\frac{2\pi}{N}k}} = \frac{1 - a^N}{1 - ae^{-j\frac{2\pi}{N}k}}$$

8.12. Suppose we have two four-point sequences $x[n]$ and $h[n]$ as follows:

$$\begin{aligned} x[n] &= \cos\left(\frac{\pi n}{2}\right), & n &= 0, 1, 2, 3, \\ h[n] &= 2^n, & n &= 0, 1, 2, 3. \end{aligned}$$

- (a) Calculate the four-point DFT $X[k]$.
- (b) Calculate the four-point DFT $H[k]$.
- (c) Calculate $y[n] = x[n] \textcircled{4} h[n]$ by doing the circular convolution directly.
- (d) Calculate $y[n]$ of part (c) by multiplying the DFTs of $x[n]$ and $h[n]$ and performing an inverse DFT.

[解]:

我们记: $W_N = e^{-j\frac{2\pi}{N}}$

(a)

这里我们只需要计算 4 点的 DFT:

$$\begin{aligned} DFT\{x[n]\} &= \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk} = X(k) = \sum_{n=0}^3 x[n] e^{-j\frac{2\pi}{N}nk} = \sum_{n=0}^3 \cos\left(\frac{\pi n}{2}\right) e^{-j\frac{\pi}{2}nk} \\ X(k) &= 1 - e^{-j\pi k} = 2j \left\{ e^{-j\frac{\pi}{2}k} \sin\left(\frac{\pi}{2}k\right) \right\} = 2je^{-j\frac{\pi}{2}k} \sin\left(\frac{\pi}{2}k\right) \quad k \in [0,3] \end{aligned}$$

答案给的形式是 $1 - W_4^{2k}$, 也没问题, 稍后我们总结什么时候需要如此表达。

(b)

$$\begin{aligned} H(k) &= \sum_{n=0}^{N-1} h[n] e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^3 2^n e^{-j\frac{\pi}{2}kn} = \sum_{n=0}^3 \left(2e^{-j\frac{\pi}{2}k}\right)^n = \frac{1 - \left(2e^{-j\frac{\pi}{2}k}\right)^{(3+1)}}{1 - 2e^{-j\frac{\pi}{2}k}} \\ &= \frac{1 - 2^4 e^{-j2\pi k}}{1 - 2e^{-j\frac{\pi}{2}k}} = \frac{-15}{1 - 2e^{-j\frac{\pi}{2}k}} \quad k \in [0,3] \end{aligned}$$

答案给的形式是 $1 + 2W_4^k + 4W_4^{2k} + 8W_4^{3k}$

(c)

题目让我们直接计算 4 点的圆卷积:

$$y[n] = x[n] \overset{4}{\otimes} h[n]$$

给出计算示意图:

$$\underset{4\text{点}}{x[n]} \overset{4}{\otimes} \underset{4\text{点}}{h[n]} = \underset{4\text{点}}{y[n]} \quad \boxed{N < N_1 + N_2 - 1} \text{有混叠}$$

$$\begin{array}{cccc|cccc} n=0 & & & & & & & \\ 1 & 2 & 4 & 8 & & & & \\ 1 & 0 & -1 & 0 & & & & \\ \hline & 0 & 0 & 0 & 0 & & & \\ -1 & -2 & -4 & -8 & & & & \\ 0 & 0 & 0 & 0 & & & & \\ \hline 1 & 2 & 4 & 8 & & & & \\ \hline 1 & 2 & 3 & 6 & -4 & -8 & 0 & \\ \uparrow & & & & & & & \\ n=0 & & & & & & & \end{array}$$

周期为4 混叠

$$\begin{array}{cccc|cccc} & & & & 1 & 2 & 3 & 6 & -4 & -8 & 0 \\ & & & & -4 & -8 & 0 & & & & \\ 1 & 2 & 3 & 6 & -4 & -8 & 0 & & & & \\ & & & & -3 & -6 & 3 & 6 & & & \end{array}$$

$$y[n] = \begin{cases} -3 & n = 0 \\ -6 & n = 1 \\ 3 & n = 2 \\ 6 & n = 3 \end{cases}$$

$$y[n] = -3 - 6\delta[n-1] + 3\delta[n-2] + 6\delta[n-3]$$

$$Y(k) = X(k)H(k) = (1 + 2W_4^k + 4W_4^{2k} + 8W_4^{3k})(1 - W_4^{2k})$$

$$Y(k) = 1 - W_4^{2k} + 2W_4^k - 2W_4^{3k} + 4W_4^{2k} - 4W_4^{4k} + 8W_4^{3k} - 8W_4^{5k}$$

$$W_4^{(n+4)k} = e^{-j\frac{2\pi}{4}(n+4)k} = e^{-j\frac{2\pi}{4}nk} = W_4^{nk}$$
$$Y(k) = -3 - 6W_4^k + 3W_4^{2k} + 6W_4^{3k}$$
$$y[n] = -3 - 6\delta[n-1] + 3\delta[n-2] + 6\delta[n-3]$$

只要涉及到圆卷积，我都建议大家用这个形式表示

8.15. Figure P8.15-1 shows two sequences $x_1[n]$ and $x_2[n]$. The value of $x_2[n]$ at time $n = 3$ is not known, but is shown as a variable a . Figure P8.15-2 shows $y[n]$, the four-point circular convolution of $x_1[n]$ and $x_2[n]$. Based on the graph of $y[n]$, can you specify a uniquely? If so, what is a ? If not, give two possible values of a that would yield the sequence $y[n]$ as shown.

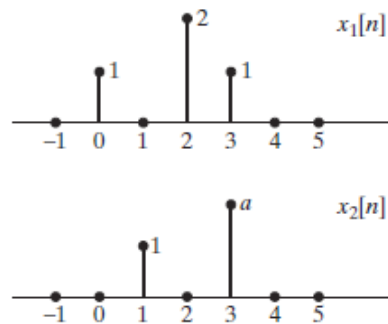


Figure P8.15-1

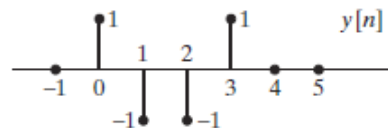


Figure P8.15-2

[解]:

$N = 4$ $N_1 = 7$ $N_2 = 7$ 所以必然会有混叠, 无法直接使用线性卷积

圆卷积是线性卷积算出来后, 对线性卷积进行周期性叠加

$$y[n] = x_1[n] \otimes_4 x_2[n] = \left\{ \sum_{N=0}^3 x_1[N] x_2[(n-N) \bmod 4] \right\} R_4[n] = \begin{cases} 1 & n=0 \\ 2a+1 & n=1 \\ a & n=2 \\ a+2 & n=3 \\ 0 & \text{otherwise} \end{cases}$$

根据数值对比, 可以确定下来, $a = -1$

8.19. Two finite-length sequences $x[n]$ and $x_1[n]$ are shown in Figure P8.19. The DFTs of these sequences, $X[k]$ and $X_1[k]$, respectively, are related by the equation

$$X_1[k] = X[k]e^{-j(2\pi km/6)},$$

where m is an unknown constant. Can you determine a value of m consistent with Figure P8.19? Is your choice of m unique? If so, justify your answer. If not, find another choice of m consistent with the information given.

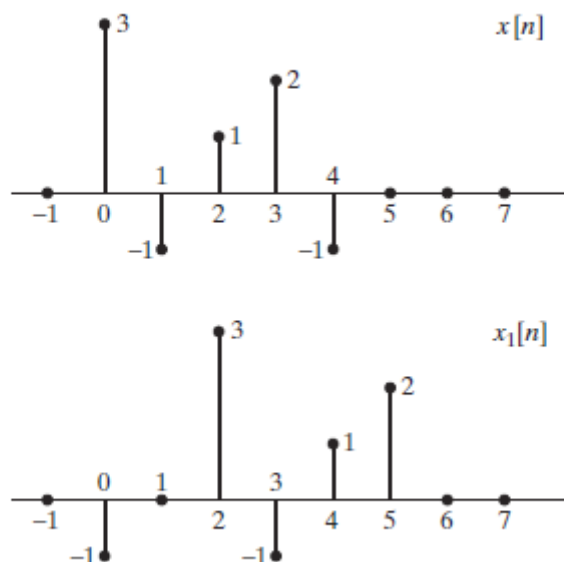


Figure P8.19

[解]:

我们先确定一个 m 对应的值:

$$X_1(k) = X(k)e^{-j\frac{2\pi}{6}km}$$

这看上去是一个时移的形式, 并且所做的 DFT 点数也是很明确的——6 点

$$W_N = W_6 = e^{-j\frac{2\pi}{6}}$$

$$X(k) = \sum_{n=0}^{6-1} x[n]W_6^{nk} = 3 - W_6^k + W_6^{2k} + 2W_6^{3k} - W_6^{4k}$$

$$X_1(k) = \sum_{n=0}^{6-1} x_1[n]W_6^{nk} = -1 + 3W_6^{2k} - W_6^{3k} + W_6^{4k} + 2W_6^{5k}$$

根据循环群的性质:

$$W_6^{(n+6)k} = e^{-j\frac{2\pi}{6}(n+6)k} = e^{-j\frac{2\pi}{6}nk} = W_6^{nk} \quad l \in N$$

$$X_1(k) = X(k)W_6^{2k}$$

所以这里有:

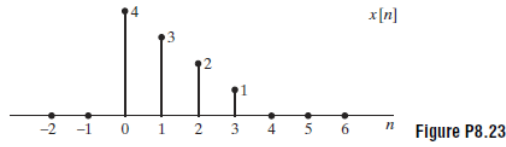
$$W_6^{2k} = e^{-j\frac{2\pi}{6}2k} = e^{-j\frac{2\pi}{6}(2+6l)k}$$

所以有 m 的通解:

$$m = 2 + 6l \quad l \in N$$

所以并不唯一确定

8.23. Consider the real finite-length sequence $x[n]$ shown in Figure P8.23.



(a) Sketch the finite-length sequence $y[n]$ whose six-point DFT is

$$Y[k] = W_6^{5k} X[k],$$

where $X[k]$ is the six-point DFT of $x[n]$.

(b) Sketch the finite-length sequence $w[n]$ whose six-point DFT is

$$W[k] = \text{Im}\{X[k]\}.$$

(c) Sketch the finite-length sequence $q[n]$ whose three-point DFT is

$$Q[k] = X[2k+1], \quad k = 0, 1, 2.$$

[解]:

(a)

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk} = \sum_{n=0}^5 x[n] e^{-j\frac{2\pi}{6}nk} = 4 + 3W_6^k + 2W_6^{2k} + W_6^{3k}$$

$$Y(k) = W_6^{5k} X(k) = 4W_6^{5k} + 3 + 2W_6^k + W_6^{2k} = 3 + W_6^k + W_6^{2k} + 4W_6^{5k}$$

$$y[n] = 3\delta[n] + \delta[n-1] + \delta[n-2] + 4\delta[n-5]$$

图略

(b)

$$W(k) = \text{Im}\{X(k)\} = \frac{1}{2j} [X(k) - X^*(k)]$$

$$X^*(k) = \left\{ \sum_{n=0}^5 x[n] e^{-j\frac{2\pi}{6}nk} \right\}^*$$

这里很明确, $x[n]$ 是必然实的, 所以会有:

$$X^*(k) = \left\{ \sum_{n=0}^5 x[n] e^{-j\frac{2\pi}{6}nk} \right\}^* = \sum_{n=0}^5 x[n] e^{+j\frac{2\pi}{6}nk} = X(-k)$$

$$W(k) = \text{Im}\{X(k)\} = \frac{1}{2j} [X(k) - X^*(k)] = \frac{1}{2j} [X(k) - X(-k)]$$

$$\begin{aligned} X(k) - X(-k) &= (4 + 3W_6^k + 2W_6^{2k} + W_6^{3k}) - (4 + 3W_6^{-k} + 2W_6^{-2k} + W_6^{-3k}) \\ &= 3W_6^k + 2W_6^{2k} - 2W_6^{4k} - 3W_6^{5k} = M(k) \end{aligned}$$

$$w[n] = \frac{1}{2j} m[n] = \frac{1}{2j} \{3\delta[n-1] + 2\delta[n-2] - 2\delta[n-4] - 3\delta[n-5]\}$$

(c)

$$Q(k) = X(2k+1) \quad k \in [0, 2]$$

$$Q(k) = 4 + 3W_6^{2k+1} + 2W_6^{2(2k+1)} + W_6^{3(2k+1)} = (4 + e^{-j\pi}) + 3e^{-j\frac{\pi}{3}}W_3^k + 2e^{-j\frac{2\pi}{3}}W_3^{2k}$$

$$= 3 + 3e^{-j\frac{\pi}{3}}W_3^k + 2e^{-j\frac{2\pi}{3}}W_3^{2k}$$

$$q[n] = 3\delta[n] + 3e^{-j\frac{\pi}{3}}\delta[n-1] + 2e^{-j\frac{2\pi}{3}}\delta[n-2]$$

8.25. Consider the signal $x[n] = \delta[n - 4] + 2\delta[n - 5] + \delta[n - 6]$.

- (a) Find $X(e^{j\omega})$ the discrete-time Fourier transform of $x[n]$. Write expressions for the magnitude and phase of $X(e^{j\omega})$, and sketch these functions.
- (b) Find all values of N for which the N -point DFT is a set of real numbers.
- (c) Can you find a three-point causal signal $x_1[n]$ (i.e., $x_1[n] = 0$ for $n < 0$ and $n > 2$) for which the three-point DFT of $x_1[n]$ is:

$$X_1[k] = |X[k]| \quad k = 0, 1, 2$$

where $X[k]$ is the three-point DFT of $x[n]$?

[解]:

(a)

$$\begin{aligned} X(e^{j\omega}) &= DTFT\{x[n]\} = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n} = e^{-j4\omega} + 2e^{-j5\omega} + e^{-j6\omega} \\ &= e^{-j5\omega}(2 + e^{+j\omega} + e^{-j\omega}) = 2e^{-j5\omega}(1 + \cos\omega) = |X(e^{j\omega})|e^{+j\angle X(e^{j\omega})} \end{aligned}$$

$$\text{Magnitude: } |X(e^{j\omega})| = 2(1 + \cos(\omega))$$

$$\text{Phase: } \angle X(e^{j\omega}) = -5\omega$$

图略

(b)

我们要做 N 点的DFT，只要可以保证 $N \geq 7$ ，也就是把 $x[n]$ 所有点都去到，那么有：

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n} = \sum_{n=0}^{N-1} x[n]e^{-j\omega n} \\ X(k) &= \left\{ \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}nk} \right\} = \left\{ \sum_{n=0}^{N-1} x[n]e^{-j\omega n} \right\} \Big|_{\omega=\frac{2\pi}{N}k} \end{aligned}$$

N 点DFT本质上是对DTFT的 N 点采样

所以如果我们要保证我们得到的DFT结果全实，那么对 $X(e^{j\omega}) = |X(e^{j\omega})|e^{+j\angle X(e^{j\omega})} = 2(1 + \cos\omega)e^{-j5\omega}$ 的采样就应该关注于 $e^{-j5\omega}$ 上：

所以我们可以取 $N = 10$

$$N = 10$$

如果我们没有 $N \geq 7$ ，会是怎么样的情景？

这个时候，我们要做DFT，就不能直接使用原有题目中给我们的 $x[n]$ 了，我们需要重构

$\tilde{x}[n] = \sum_{r=-\infty}^{+\infty} x[n - rN]$ ，然后对 $\tilde{x}[n]$ 做DFS：

$$\begin{aligned} DFS\{\tilde{x}[n]\} &= \tilde{X}(k) = \sum_{n=0}^{N-1} \tilde{x}[n]e^{-j\frac{2\pi}{N}nk} \\ X(k) &= \begin{cases} \tilde{X}(k) & 0 \leq k \leq N-1 \\ 0 & \text{otherwise} \end{cases} \\ \tilde{X}(k) &= X(e^{j\omega}) \Big|_{\omega=\frac{2\pi}{N}k} \end{aligned}$$

这个地方我们有 $\tilde{X}(k)$ 全是实数，那么自然在主周期上会有 $X(k)$ 全实数

所以还会有：

$$N = 1, 2, 5$$

于是有：

$$N = 1, 2, 5, 10$$

(c)

先做出来 $x[n]$ 的三点DFT，这里我们先要做出：

$$\tilde{x}[n] = \sum_{r=-\infty}^{+\infty} x[n - 3r]$$

在做DFS：

$$\tilde{X}(k) = \sum_{n=0}^{3-1} \tilde{x}[n] e^{-j\frac{2\pi}{3}nk}$$

最后在主周期上做限制：

$$X(k) = \tilde{X}(k)R_3(k) = \begin{cases} \tilde{X}(k) & 0 \leq k \leq 2 \\ 0 & otherwise \end{cases}$$

在 $n \in [0, 2]$ 上有：

$$\begin{aligned} \tilde{x}[n] &= \delta[n] + \delta[n - 1] + 2\delta[n - 2] \\ \tilde{X}(k) &= \sum_{n=0}^{3-1} \tilde{x}[n] e^{-j\frac{2\pi}{3}nk} = 1 + e^{-j\frac{2\pi}{3}k} + 2e^{-j\frac{4\pi}{3}k} \end{aligned}$$

这里我们只提取主周期 $k \in [0, 2]$ ，不提取其他范围，让其他范围为0，就会有：

$$\begin{aligned} X(k) &= 1 + e^{-j\frac{2\pi}{3}k} + 2e^{-j\frac{4\pi}{3}k} \quad k \in [0, 2] \\ X(0) &= 4 \quad X(1) = -\frac{1}{2} + \frac{\sqrt{3}}{2}j \quad X(2) = -\frac{1}{2} - \frac{\sqrt{3}}{2}j \\ |X(0)| &= 4 \quad |X(1)| = 1 \quad |X(2)| = 1 \end{aligned}$$

$$X_1(k) = \sum_{n=0}^2 x_3(n) e^{-j\frac{2\pi}{3}nk}$$

不妨记 $x_1[n] = a\delta[n] + b\delta[n - 1] + c\delta[n - 2]$

$$\begin{cases} a + b + c = 4 \\ a + be^{-j\frac{2\pi}{3}} + ce^{-j\frac{4\pi}{3}} = 1 \\ a + be^{-j\frac{4\pi}{3}} + ce^{-j\frac{8\pi}{3}} = 1 \end{cases}$$

可以得到：

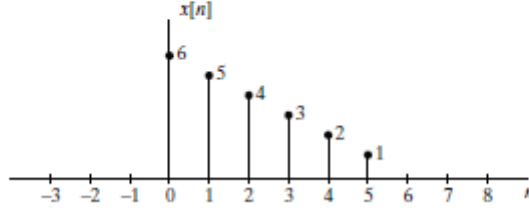
$$\begin{cases} a = 2 \\ b = 1 \\ c = 1 \end{cases}$$

$$x_1[n] = 2\delta[n] + \delta[n - 1] + \delta[n - 2]$$

8.28. Consider the six-point sequence

$$x[n] = 6\delta[n] + 5\delta[n-1] + 4\delta[n-2] + 3\delta[n-3] + 2\delta[n-4] + \delta[n-5]$$

shown in Figure P8.28.



- Determine $X[k]$, the six-point DFT of $x[n]$. Express your answer in terms of $W_6 = e^{-j2\pi/6}$.
- Plot the sequence $w[n]$, $n = 0, 1, \dots, 5$, that is obtained by computing the inverse six-point DFT of $W[k] = W_6^{-2k} X[k]$.
- Use any convenient method to evaluate the six-point circular convolution of $x[n]$ with the sequence $h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$. Sketch the result.
- If we convolve the given $x[n]$ with the given $h[n]$ by N -point circular convolution, how should N be chosen so that the result of the circular convolution is identical to the result of linear convolution? That is, choose N so that

$$\begin{aligned} y_p[n] &= x[n] \circledast h[n] = \sum_{m=0}^{N-1} x[m]h[(n-m)_N] \\ &= x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m]h[n-m] \quad \text{for } 0 \leq n \leq N-1. \end{aligned}$$

- In certain applications, such as multicarrier communication systems (see Starr et al, 1999), the linear convolution of a finite-length signal $x[n]$ of length L samples with a shorter finite-length impulse response $h[n]$ is required to be identical (over $0 \leq n \leq L-1$) to what would have been obtained by L -point circular convolution of $x[n]$ with $h[n]$. This can be achieved by augmenting the sequence $x[n]$ appropriately. Starting with the graph of Figure P8.28, where $L = 6$, add samples to the given sequence $x[n]$ to produce a new sequence $x_1[n]$ such that with the sequence $h[n]$ given in part (c), the ordinary convolution $y_1[n] = x_1[n] * h[n]$ satisfies the equation

$$\begin{aligned} y_1[n] &= x_1[n] * h[n] = \sum_{m=-\infty}^{\infty} x_1[m]h[n-m] \\ &= y_p[n] = x[n] \circledast h[n] = \sum_{m=0}^5 x[m]h[(n-m)_6] \quad \text{for } 0 \leq n \leq 5. \end{aligned}$$

- Generalize the result of part (c) for the case where $h[n]$ is nonzero for $0 \leq n \leq M$ and $x[n]$ is nonzero for $0 \leq n \leq L-1$, where $M < L$; i.e., show how to construct a sequence $x_1[n]$ from $x[n]$ such that the linear convolution $x_1[n] * h[n]$ is equal to the circular convolution $x[n] \circledast h[n]$ for $0 \leq n \leq L-1$.

(a).

$$X(k) = \sum_{n=0}^5 x[n]e^{-j\frac{\pi}{3}kn} = 6 + 5W_6^k + 4W_6^{2k} + 3W_6^{3k} + 2W_6^{4k} + W_6^{5k}$$

(b).

$$\begin{aligned} W(k) &= W_6^{-2k} X(k) = 6W_6^{-2k} + 5W_6^{-k} + 4 + 3W_6^k + 2W_6^{2k} + W_6^{3k} = 4 + 3W_6^k + 2W_6^{2k} + W_6^{3k} + 6W_6^{4k} + 5W_6^{5k} \\ w[n] &= \frac{1}{N} \sum_{k=0}^5 W(k)e^{+j\frac{\pi}{3}kn} = 4\delta[n] + 3\delta[n-1] + 2\delta[n-2] + \delta[n-3] + 6\delta[n-4] + 5\delta[n-5] \end{aligned}$$

(c).

$$\begin{aligned} x[n] \circledast h[n] &= s[n] \\ x[n] &= 6\delta[n] + 5\delta[n-1] + 4\delta[n-2] + 3\delta[n-3] + 2\delta[n-4] + \delta[n-5] \end{aligned}$$

$$h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

$$s[n] = \left\{ \sum_{m=0}^5 x[m] h[(n-m)_6] \right\} R_6[n] = \left\{ \sum_{m=0}^5 x[m] \sum_{r=-\infty}^{+\infty} h[n-m+6r] \right\} R_6[n] = \left\{ \sum_{r=-\infty}^{+\infty} \sum_{m=0}^5 x[m] h[n-m+6r] \right\} R_6[n]$$

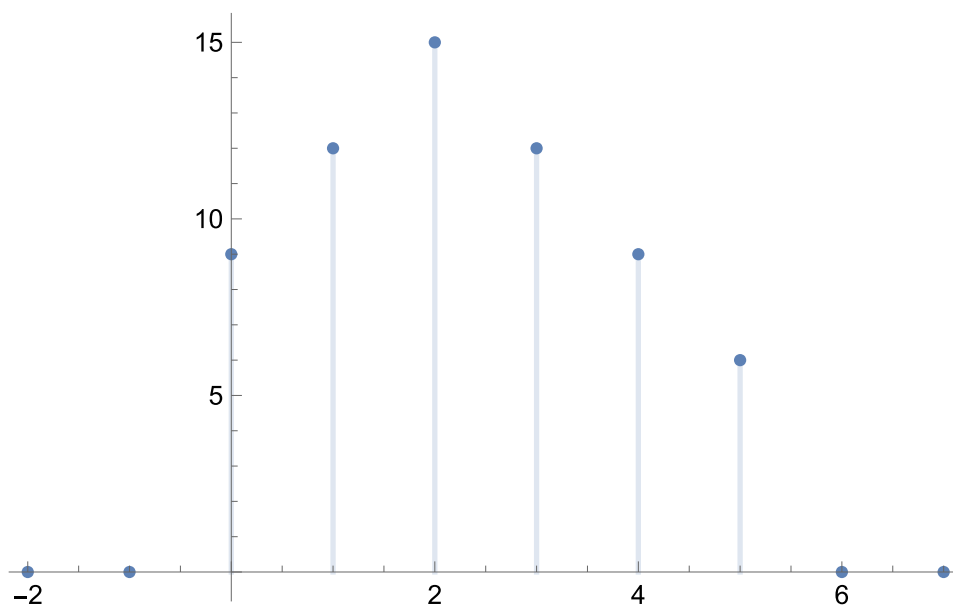
$$\text{上式} = \left\{ \sum_{r=-\infty}^{+\infty} g[n+6r] \right\} R_6[n]$$

$$g[n] = 6\delta[n] + 11\delta[n-1] + 15\delta[n-2] + 12\delta[n-3] + 9\delta[n-4] + 6\delta[n-5] + 3\delta[n-6] + \delta[n-7]$$

于是圆卷积：

$$s[n] = 9\delta[n] + 12\delta[n-1] + 15\delta[n-2] + 12\delta[n-3] + 9\delta[n-4] + 6\delta[n-5]$$

结果见下图：



(d).

选取 N 满足：

$$N \geq N_1 + N_2 - 1$$

即线性卷积做搬移的时候没有 aliasing（交叠）

(e).

$$h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

增加点数得到：

$$x_1[n] = 2\delta[n+2] + \delta[n+1] + 6\delta[n] + 5\delta[n-1] + 4\delta[n-2] + 3\delta[n-3] + 2\delta[n-4] + \delta[n-5]$$

(f). 将 $x[n]$ 的右边高位的 $S = \{(L-1) + (M+1) - 1\} - L = M - 1$ 项移到 $n = 0$ 左端即可

8.29. Consider the real five-point sequence

$$x[n] = \delta[n] + \delta[n-1] + \delta[n-2] - \delta[n-3] + \delta[n-4].$$

The deterministic autocorrelation of this sequence is the inverse DTFT of

$$C(e^{j\omega}) = X(e^{j\omega})X^*(e^{j\omega}) = |X(e^{j\omega})|^2,$$

where $X^*(e^{j\omega})$ is the complex conjugate of $X(e^{j\omega})$. For the given $x[n]$, the autocorrelation can be found to be

$$c[n] = x[n] * x[-n].$$

- (a) Plot the sequence $c[n]$. Observe that $c[-n] = c[n]$ for all n .
- (b) Now assume that we compute the seven-point DFT ($N = 5$) of the sequence $x[n]$. Call this DFT $X_5[k]$. Then, we compute the inverse DFT of $C_5[k] = X_5[k]X_5^*[k]$. Plot the resulting sequence $c_5[n]$. How is $c_5[n]$ related to $c[n]$ from part (a)?
- (c) Now assume that we compute the 10-point DFT ($N = 10$) of the sequence $x[n]$. Call this DFT $X_{10}[k]$. Then, we compute the inverse DFT of $C_{10}[k] = X_{10}[k]X_{10}^*[k]$. Plot the resulting sequence $c_{10}[n]$.
- (d) Now suppose that we use $X_{10}[k]$ to form $D_{10}[k] = W_{10}^{5k}C_{10}[k] = W_{10}^{5k}X_{10}[k]X_{10}^*[k]$, where $W_{10} = e^{-j(2\pi/10)}$. Then, we compute the inverse DFT of $D_{10}[k]$. Plot the resulting sequence $d_{10}[n]$.

[解]:

因为这里 $x[n]$ 是实的, 所以会有:

$$DTFT\{x[n]\} = X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n}$$

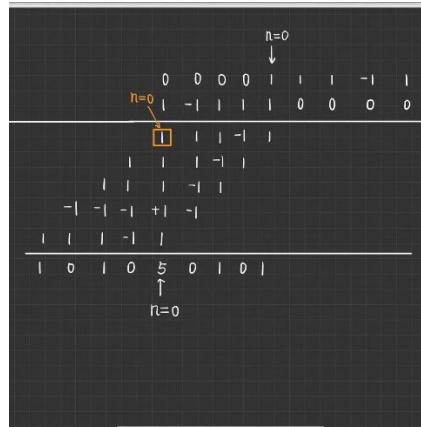
$$X^*(e^{j\omega}) = \left\{ \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n} \right\}^* = \sum_{n=-\infty}^{+\infty} x[n]e^{+j\omega n} = \sum_{n=-\infty}^{+\infty} x[n]e^{-j(-\omega)n} = X(e^{j(-\omega)})$$

$$X(e^{j(-\omega)}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j(-\omega)n} = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega(-n)} = \sum_{n=-\infty}^{+\infty} x[-n]e^{-j\omega n} = DTFT\{x[-n]\}$$

(a)

$$C(e^{j\omega}) = X(e^{j\omega})X^*(e^{j\omega}) = |X(e^{j\omega})|^2$$

$$c[n] = x[n] * x[-n]$$



$$c[n] = \delta[n+4] + \delta[n+2] + 5\delta[n] + \delta[n-2] + \delta[n-4]$$

(b)

这里题目的意思应该是，让我们先对 $x[n]$ 做 5 点DFT，然后再对 $C(k) = X(k)X^*(k)$ 做 5 点IDFT

我们有：

$$\begin{aligned}\tilde{x}[n] &= \sum_{r=-\infty}^{+\infty} x[n - rN] = \sum_{r=-\infty}^{+\infty} x[n - 5r] \\ \tilde{X}[k] &= \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\frac{2\pi}{N}nk} = \sum_{n=0}^4 \tilde{x}[n] e^{-j\frac{2\pi}{5}nk} \\ X(k) &= \tilde{X}(k)R_N(k) = \tilde{X}(k)R_5(k)\end{aligned}$$

这里经过比对就会有：

$$X(k) = \sum_{n=0}^4 x[n] e^{-j\frac{2\pi}{5}nk}$$

那么会有：

$$X^*(k) = \left\{ \sum_{n=0}^4 x[n] e^{-j\frac{2\pi}{5}nk} \right\}^* = \sum_{n=0}^4 x[n] e^{-j\frac{2\pi}{5}(-n)k} = \sum_{n=-4}^0 x[-n] e^{-j\frac{2\pi}{5}(-n)k} = DFT\{x[-n]\}$$

所以会有：

$$\begin{aligned}c_5[n] &= x[n] \overset{5}{\otimes} x[-n] \\ X_5(k) &= 1 + W_5^k + W_5^{2k} - W_5^{3k} + W_5^{4k} \\ C_5(k) &= X_5(k)X_5^*(k) = (1 + W_5^k + W_5^{2k} - W_5^{3k} + W_5^{4k})(1 + W_5^k + W_5^{2k} - W_5^{3k} + W_5^{4k})^* \\ C_5(k) &= 5 + W_5^k + W_5^{2k} + W_5^{3k} + W_5^{4k} \\ c_5[n] &= 5\delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \delta[n-4]\end{aligned}$$

$c_5[n]$ 是圆卷积结果，是(a)中线性卷积周期性搬移(周期为 5)后存在混叠的结果

$$c_5[n] = \sum_{r=-\infty}^{+\infty} c[n - rN] = \sum_{r=-\infty}^{+\infty} c[n - 5r]$$

(c)

这里 $c_{10}[n]$ 也会是(a)中线性卷积周期搬移后的结果，并且周期为 10，但是这里因为搬移周期 10 大于线性卷积得到的序列长度 9，所以不存在混叠，我们可以直接得到：

$$\begin{aligned}c_{10}[n] &= 5\delta[n] + \delta[n-2] + \delta[n-4] + \delta[n-6] + \delta[n-8] \\ c_{10}[n] &= \sum_{r=-\infty}^{+\infty} c[n - rN] = \sum_{r=-\infty}^{+\infty} c[n - 10r]\end{aligned}$$

(d)

$$\begin{aligned}D_{10}(k) &= W_{10}^{5k} C_{10}(k) \\ C_{10}(k) &= 5W_{10}^0 + W_{10}^{2k} + W_{10}^{4k} + W_{10}^{6k} + W_{10}^{8k} \\ D_{10}^k(k) &= 5W_{10}^{5k} + W_{10}^{7k} + W_{10}^{9k} + W_{10}^{11k} + W_{10}^{13k} = 5W_{10}^{5k} + W_{10}^{7k} + W_{10}^{9k} + W_{10}^{1k} + W_{10}^{3k} \\ d_{10}[n] &= \delta[n-1] + \delta[n-3] + 5\delta[n-5] + \delta[n-7] + \delta[n-9]\end{aligned}$$

8.42. $x[n]$ is a finite-length sequence of length 1024, i.e.,

$$x[n] = 0 \text{ for } n < 0, n > 1023.$$

The autocorrelation of $x[n]$ is defined as

$$c_{xx}[m] = \sum_{n=-\infty}^{\infty} x[n]x[n+m],$$

and $X_N[k]$ is defined as the N -point DFT of $x[n]$, with $N \geq 1024$.

We are interested in computing $c_{xx}[m]$. A proposed procedure begins by first computing the N -point inverse DFT of $|X_N[k]|^2$ to obtain an N -point sequence $g_N[n]$, i.e.,

$$g_N[n] = N\text{-point IDFT} \left\{ |X_N[k]|^2 \right\}.$$

- (a) Determine the minimum value of N so that $c_{xx}[m]$ can be obtained from $g_N[n]$. Also specify how you would obtain $c_{xx}[m]$ from $g_N[n]$.
- (b) Determine the minimum value of N so that $c_{xx}[m]$ for $|m| \leq 10$ can be obtained from $g_N[n]$. Also specify how you would obtain these values from $g_N[n]$.

[解]:

这里 *Autocorrelation* 叫做自相关, 在通信和一切与 *Fourier* 有关的内容中会被广泛使用, 特别是和信号能量、功率这两个量相关的事情中, 这个形式会被广泛使用

$$c_{xx}[m] = \sum_{n=-\infty}^{+\infty} x[n]x[n+m] \text{ 是一个关于变量 } m \text{ 的值}$$

我们可以很轻松判断出来, $m \geq 1024$ or $m \leq -1024$ 时, 就有:

$$c_{xx}[m] = 0$$

所以 $c_{xx}[m]$ 可能非 0, 只会在 $m \in [-1023, 1023]$ 范围内成立, 我们可以做 $N \geq 2047$ 点的 *DFT*, 大家就知道是怎么回事了。题目中需要最小的 N , 我们不妨令 $N = 2047$:

$$\tilde{c}_{xx}[m] = \sum_{r=-\infty}^{+\infty} c_{xx}[m - rN]$$

$$\tilde{C}_{xx}(k) = \sum_{m=0}^{N-1} \tilde{c}_{xx}[m] e^{-j\frac{2\pi}{N}km}$$

$$C_{xx}(k) = \tilde{C}_{xx}(k) R_N(k) = \sum_{m=0}^{N-1} \tilde{c}_{xx}[m] e^{-j\frac{2\pi}{N}km}$$

$$\sum_{m=0}^{N-1} \tilde{c}_{xx}[m] e^{-j\frac{2\pi}{N}km} = \sum_{m=0}^{N-1} \left(\sum_{n=-\infty}^{+\infty} x[n]x[n+m] \right) e^{-j\frac{2\pi}{N}km} = \sum_{n=-\infty}^{+\infty} x[n] \sum_{m=0}^{N-1} x[n+m] e^{-j\frac{2\pi}{N}km}$$

$$\begin{aligned} \sum_{n=-\infty}^{+\infty} x[n] \sum_{m=0}^{N-1} x[n+m] e^{-j\frac{2\pi}{N}km} &= \sum_{n=-\infty}^{+\infty} x[n] \left(\sum_{m=0}^{N-1} x[n+m] e^{-j\frac{2\pi}{N}km} \right) \\ &= \sum_{n=-\infty}^{+\infty} x[n] e^{+j\frac{2\pi}{N}kn} \left(\sum_{m=0}^{N-1} x[n+m] e^{-j\frac{2\pi}{N}k(m+n)} \right) \end{aligned}$$

$$\sum_{m=0}^{N-1} x[n+m] e^{-j\frac{2\pi}{N}k(m+n)} = \sum_{m'=n}^{N-1+n} x[m'] e^{-j\frac{2\pi}{N}k(m')}$$

对于这个式子大家可以画图观察, 它本质上就是 2047 点的 $x[n]$ 的 *DFT*。

$$\sum_{m=0}^{N-1} x[n+m] e^{-j\frac{2\pi}{N}k(m+n)} = \sum_{m'=n}^{N-1+n} x[m'] e^{-j\frac{2\pi}{N}k(m')} = X_N(k) = X_{2047}(k)$$

$$\sum_{n=-\infty}^{+\infty} x[n]e^{+j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} x[n]e^{+j\frac{2\pi}{N}kn} = X_N^*(k) = X_{2047}^*(k)$$

所以会有：

$$C_{xx}(k) = |X_{2047}(k)|^2$$

做2047点的反变换IDFT:

$$IDFT\{C_{xx}(k)\} = IDFT\{|X_{2047}(k)|^2\}$$

会有：

$$c_{xx}[m] = \begin{cases} g_{2047}[m] & 0 \leq m \leq 1023 \\ g[m + 2047] & -1023 \leq m < 0 \end{cases}$$

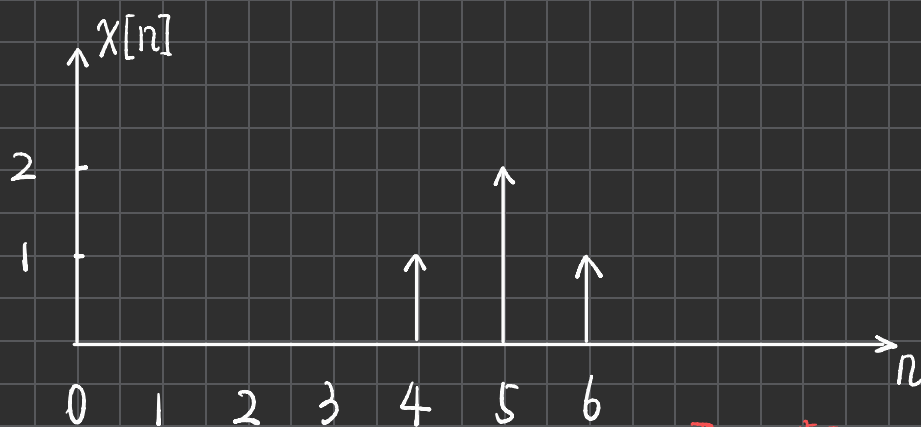
(b)

这里只要求 $|m| \leq 10$ 可以正常求就好了。所以只要这个区域不混叠，能够正常地求取即可，其他区域混叠了也无所谓。那取 $N = 1034$ 即可，因为 $m = -1024$ 以及 $m = +1024$ 全为0，纵使搬移混叠也无法影响到端点 $m = +10$ 和 $m = -10$

事实上, DFT 与 DFT 的关系并非是采样那么简单
谈论二者关系时, 我们首先要明确如何做 DFT

约定: $W_N = e^{-j \cdot \frac{2\pi}{N}} = \exp(-j \cdot \frac{2\pi}{N})$

Q: 请做 $x[n] = \delta[n-4] + 2\delta[n-5] + \delta[n-6]$ 的 3 点 DFT



T8.25 涉及

请问是: $X(k) = \sum_{n=0}^{3-1} x[n] W_3^{n \cdot k}$ $k \in [0, 3-1]$ 么?
大家动手算

$$X(0) = 0$$

$$X(1) = 0$$

$$X(2) = 0$$

(?)

事实上 DFT 不是 $\sum_{n=0}^{N-1} x[n] W_N^{nk} = X(k)$

对于 $x[n]$, 要做 N 点 DFT, 先要构造:

$$\tilde{x}[n] = \sum_{k=-\infty}^{+\infty} x[n-k \cdot N]$$

求 DFS	$\tilde{X}(k) = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{nk}$	$k \in N$
求 DFT	$X(k) = \tilde{X}(k) R_N(k) = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{nk}$	$k \in [0, N-1]$

$$\tilde{X}(k) = \sum_{n=0}^{N-1} \left(\sum_{k=-\infty}^{+\infty} x[n-k \cdot N] \right) e^{-j \frac{2\pi}{N} \cdot n \cdot k}$$

求和换序

$$= \sum_{k=-\infty}^{+\infty} \left(\sum_{n=0}^{N-1} x[n-kN] e^{-j \frac{2\pi}{N} k \cdot n} \right)$$

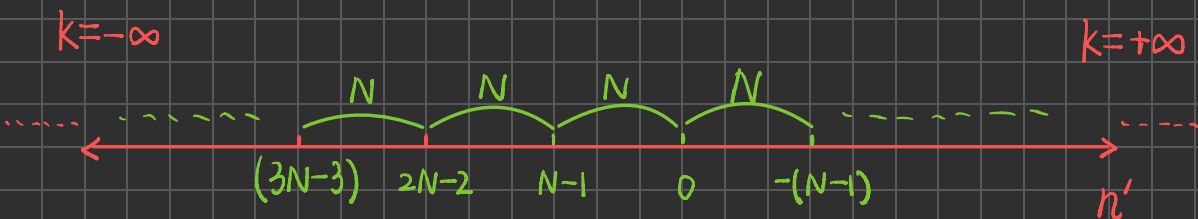
换元 $n' = n - kN$

$$= \sum_{k=-\infty}^{+\infty} \left(\sum_{n'=-kN}^{N-1-kN} x[n'] e^{-j \frac{2\pi}{N} k \cdot (n' + kN)} \right)$$

$$= \sum_{k=-\infty}^{+\infty} \sum_{n'=-kN}^{N-1-kN} x[n'] e^{-j \frac{2\pi}{N} k \cdot n'}$$

首尾相连

$$= \sum_{n'=-\infty}^{+\infty} x[n'] e^{-j \frac{2\pi}{N} k \cdot n'}$$



更换上下限

$$= \sum_{n'=-\infty}^{+\infty} x[n'] e^{-j \cdot \frac{2\pi}{N} \cdot k \cdot n'}$$

即: $\tilde{X}(k) = \sum_{n'=-\infty}^{+\infty} x[n'] e^{-j \frac{2\pi}{N} k n'} = X(e^{j\omega}) \Big|_{\omega = \frac{2\pi}{N} \cdot k}$

DFS是DTFT的采样

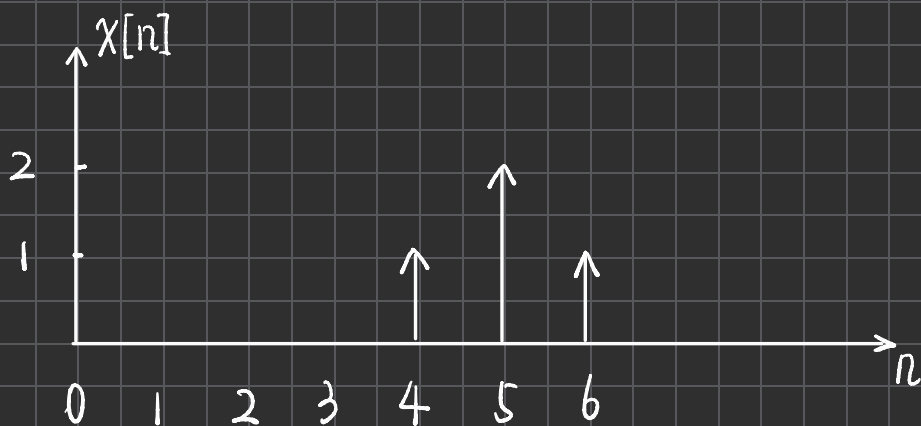
事实上 DFT 对应 $X(k)$ $k \in [0, N-1]$

DTFT 对应 $X(e^{j\omega})$ 周期为 2π

DFS 对应 $\tilde{X}(k)$ 周期为 N

则可能是 DFS 对应 DFT 采样

所以对上方的问题:



↓ 延拓



$$\text{DFS: } \tilde{X}(k) = \sum_{n=0}^{3-1} \tilde{x}[n] W_3^{nk} = 1 + W_3^k + 2W_3^{2k} \quad k \in \mathbb{N}$$

$$\text{DFT: } X(k) = \tilde{X}(k) R_3(k) = 1 + W_3^k + 2W_3^{2k} \quad k \in [0, 2]$$